Explainable Machine Learning Framework For Early Colorectal Cance Risk Prediction Using Gut microbiame and life style data

21CSP302L - M I N O R PROJECT

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ABSTRACT

Cancer of the colon or rectum is regarded as one of the deadliest cancers that exist in the whole world. The rising rates of cancer are largely attributed to poor diets, being inactive, being overweight, and alterations in the gut bacteria. Tests and scans used by physicians to diagnose these diseases come with high charges as well as long periods and invasion of the human body. They detect the cancer in its advanced stages thus rendering treatment very hard.

The early signs of colon or rectal cancer can be detected through examining the bacteria existing in the gut and examining how active people are in their everyday life without the involvement of any invasive process. Gut bacteria reveal details which should be coupled with what a person eats, how active he/she is, whether he/she smokes, and his/her body weight. This becomes a mechanism of detecting risks early enough.

The objective of this research study is to design an explainable machine learning architecture that would help in determining the early risk prediction of colorectal cancer using the microbiome composition of the gut as well as certain lifestyle factors. This framework uses the available public data for microbiome along with other lifestyle variables to determine the risk factor for early stages of cancer.

Incorporating XAI approaches like SHAP (SHapley Additive explanations) is intended to assist in interpreting the decisions of the machine learning algorithm while identifying significant factors affecting cancer risk prediction. Based on the risk prediction, the individual is classified into one of three categories: low risk, medium risk, or high risk.

This model has achieved an 81.73% recall rate and an 84.22% F1-score, thus proving the capability of detecting patients who are at risk of getting colon cancer. This model surpasses other models because it offers clarity and is also non-invasive and timely.

Key Words: Colon Cancer, Gut Microbiome, Machine Learning, Explainable AI, SHAP, Random Forest,LogisticRegression

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ABBREVIATIONS

AI – Artificial Intelligence
ML – Machine Learning
XAI – Explainable Artificial Intelligence
SHAP – shapley Additive explanations
CRC – Colorectal Cancer
BMI – Body Mass Index
LR – Logistic Regression
RF – Random Forest
CLR – Centered Log Ratio
TP – True Positive
TN – True Negative
FP – False Positive
FN – False Negative
ROC – Receiver Operating Characteristic
AUC – Area Under Curve

CHAPTER 1

INTRODUCTION

1.1 Introduction to Project

Colorectal cancer (CRC) is among the top reasons for deaths caused by cancer globally. Early detection of the disease improves the chances of survival and makes treatment easier. Traditional diagnosis methods such as colonoscopy, biopsy, and imaging are highly invasive, expensive, and tend to diagnose CRC in advanced stages.

The current revolution in AI and machine learning allows developing highly efficient systems capable of analyzing complex biomedical information. Recent findings showed that gut microbiota is highly related to the risks of CRC because it represents a good predictor of the illness development [7],[8].

This proposed system would entail developing an Explainable Machine Learning-based Colorectal Cancer Risk Prediction System by taking into account the composition of the gut microbiome and patients' lifestyles such as their diets, body mass index, smoking habits, and physical activities. Data preprocessing involves data normalization, data encoding, and feature integration before applying logistic regression and random forest classifiers.

The explanation of machine learning models enables the discovery of the contribution of each feature, thereby enhancing the interpretability of results through the use of the SHAP approach. The current research concentrates on creating a better way to detect colorectal cancer risk.

1.2 Problem Statement

The diagnosis of colorectal cancer is one of the significant challenges faced by modern healthcare because of delayed detection and higher frequency. Currently used methods for colorectal cancer diagnosis are invasive and depend upon the expertise of medical practitioners.

There are a few problems associated with existing solutions:

- Late detection of cancerous changes: Current approaches do not help detect cancer at an early stage.
- Invasive methods: Imaging and colonoscopy are expensive and painful.
- Insufficient amount of data used in predictions: Only genetic information or images are utilized; other factors are not considered.
- Inefficiency: AI systems often function as black-box models.

These shortcomings result in the need for an innovative and understandable solution capable of analyzing both the micro biome and lifestyles data. The necessity for such a method has been emphasized recently in the works that have analyzed machine learning-based cancer prediction systems [2],[4],[6].

1.3 Motivation

The motivation for this project is to solve problems with existing techniques for colorectal cancer detection, using machine learning and explainable artificial intelligence techniques. Existing techniques only analyze either the microbiome data or the lifestyle data, but do not take into account their combination.

Recent studies have shown that using machine learning on microbiome data results in significantly improved prediction performance and early detection [7],[8]. However, most solutions are difficult to interpret and applicable in real world situations.

The goal of this project is to create an integrated solution that involves:

Microbiome analysis
Analysis of lifestyle data
Prediction using machine learning techniques
Explainable AI

This solution will help meet modern-day needs in health care by making early, non-invasive and interpretable predictions.

1.4 Sustainable Development Goal of The Project

The current project is closely related to **SDG 3: Good Health and Well-being**. This goal aims to ensure healthy lives and promote well-being for all people at all ages. Within this goal, there is an important target focused on minimizing premature mortality caused by non-communicable diseases (including cancer) through early detection, prevention, and effective treatment measures.

Colorectal cancer is considered one of the most common non-communicable diseases. Because of the lack of effective screening methods, the disease tends to be diagnosed in late stages, leading to ineffective treatment and low chances of survival. The design of a non-invasive, data-driven model for early risk evaluation of colorectal cancer would greatly contribute to resolving the problem.

The application of machine learning allows for the efficient examination of biological and behavioral factors, facilitating early risk prediction. In addition, the role of XAI in this scenario cannot be overestimated. Application of the suggested technology is critical for practical deployment within health care because of its transparency and accountability.

CHAPTER 2

2. LITERATURE SURVEY

2.1 Overview of The Research Area

Currently, there has been growing attention to the application of data-driven approaches in the field of medicine, with respect to such an issue as disease prediction and prevention. There are increasing efforts made regarding the study of colon cancer, which is the leading cause of cancer-related mortality in the world. Normally, the use of diagnostic visualization methods such as colonoscopy, computed tomography, or histopathology is used for diagnosing this condition. These approaches have proven to be very effective; nevertheless, they can only be used for detecting the disease.Meanwhile, biological studies of recent times indicate the important role played by the gut microbiome in human health and diseases. Alterations in the structure and variety of intestinal bacteria are regarded as indicators of cancer at its early stage of development. Thus, there appeared a whole line of research dedicated to finding patterns and markers in microbiome data to be used in cancer risk prediction.

It should be mentioned that certain lifestyle factors, like nutrition, smoking, drinking alcohol, exercise, and body mass index, also influence the likelihood of developing colorectal cancer. Therefore, contemporary scientists tend to take into account not only biological factors but also those related to lifestyles of people.

Nevertheless, because of the development of machine learning and artificial intelligence methods, the investigators can use large and sophisticated databases to uncover some latent correlations between certain risk factors. Examples of such machine learning algorithms are Logistic Regression, Random Forest, and Gradient Boosting, which were implemented in practice for classification and prediction in the health care area. The recent ten years were marked by the growth of interest in the realm of Explainable Artificial Intelligence (XAI) as it allows developing explanations on the basis of the achieved results.

To conclude, today the investigation area is shifting from the diagnostic approach to the predictive one.

2.2 Existing Models and Frameworks

There has been considerable advancement made recently in detecting and analyzing colorectal cancers using both traditional and computational methods. Some of these advancements can be classified as those involving imaging techniques, microbiome analysis techniques, and those based on lifestyle factors.

Imaging-Based Models

It is pertinent to note that a substantial amount of literature on the topic is devoted to the analysis of medical images using colonoscopy and histopathology. The use of deep learning algorithms, especially CNNs, has become popular in detecting any abnormalities, including polyps and tumors. Models like ResNet, VGG, and Inception networks have proven to be very accurate in the identification of cancerous areas. However, although these models work efficiently, their scope is limited to the detection of cancer after its appearance

.**Microbiome-Based Models**

One of the commonest research areas involves the analysis of colonoscopy or histopathology images. Deep learning models, mainly convolutional neural networks, have been frequently used for the identification of abnormalities such as polyps or tumors. Models like ResNet, VGG, and Inception networks have been highly effective in locating cancer in these images. However, all these models are basically used only in post-diagnosis situations where there is a need to identify the cancer after its occurrence. Moreover, the implementation of such models needs expensive and dedicated machinery and environment.

Lifestyle-Based Models

There have been several studies conducted on the prediction of cancer from gut microbiome data. Various machine learning models like random forest, SVM, and clustering models have been found highly effective for predicting colorectal cancer from biological data. Despite their effectiveness, these models mostly work on biological data and ignore several other factors related to lifestyle. There are very few works on explaining how these models function

Hybrid and Multimodal Approaches

There are a few recent works where the authors tried to integrate various types of input data like genes and lifestyle factors into models for more efficient predictions. Multimodal techniques show better results than those based on a single type of input data. Still, most of these models suffer from some problems like insufficient data integration capabilities, scalability issues, and absence of interpretability.

Limitations of Existing Frameworks

Even though there are some advances in the field, existing models have a lot of weaknesses:

- Majority of them concentrate on detection rather than prediction
- Low involvement of microbiome and lifestyle data
- High reliance on medical and imaging data
- No interpretability of machine learning algorithms
- Few options for use in non-invasive screenings

These problems indicate the necessity to create an innovative model that will incorporate multiple risk factors and provide accurate and explainable prediction results.

2.3 Limitations Identified from Literature Survey

Firstly, most of these approaches concentrate on detecting a certain disease after its onset or development. The accuracy rates of the imaging-based models are high; however, these models can detect colorectal cancer only when tumors or other anomalies have already been formed.

Secondly, there is no application of multiple types of data within one model. Each existing algorithm uses a particular data source, whether it is imaging, genetics, or microbiome information. Colorectal cancer risk depends not only on biological characteristics but also on patients' lifestyles. Hence, using different data types will provide more precise prediction results.

Moreover, the algorithms and models created by researchers belong to black box models, and there is a need for an explanation of a particular prediction result. In healthcare, this aspect poses serious difficulties for practitioners who need to understand the reasoning behind the prediction.

Finally, there is a heavy reliance on the presence of clinical infrastructure. Methods such as imaging and laboratory testing require special equipment and skilled experts to perform analyses.

Additionally, there is a problem with variability and inconsistency in the data. For instance, the microbiome data varies widely from one individual to another depending on their geographical location, diet, and living conditions, which makes it hard to develop general models.

Finally, current systems are not scalable and have limited practical applications since they are designed for experimental settings.

2.4 Research Objectives

The most fundamental objective of the present project is to create a system that enables intelligent and explainable prediction of the risk of getting colorectal cancer based on non-invasive sources of information. In order to achieve the goal, specific objectives have been outlined that should be accomplished during the development of the system.

Some of the important objectives of the present project are as follows:

- To design a machine learning-based model for prediction of the risk of developing colorectal cancer
- To obtain and analyze information about the gut microbiome and related variables like diet, smoking, exercise, and body mass index
- To combine information from various sources to obtain a single data set
- To implement and compare various machine learning algorithms such as logistic regression and random forest classifier
- To conduct performance evaluation of models using parameters like accuracy, precision, recall, and ROC-AUC
- To incorporate techniques of explainable artificial intelligence (AI) like SHAP into model building process
- To classify people based on their risk as Low, Moderate, and High

2.5 Product backlog

Product Backlog Preparation

The product backlog in this case is developed according to the Agile development methodology, where development activities are decomposed into smaller units called user stories. A user story is defined by the functionality or requirements needed to develop the whole system. The product backlog will be sorted in order to implement crucial features initially and then improve them incrementally over various sprints.

For the current project, the product backlog involves creating a full pipeline from data processing to model training, implementing explainable AI, and deploying the system.

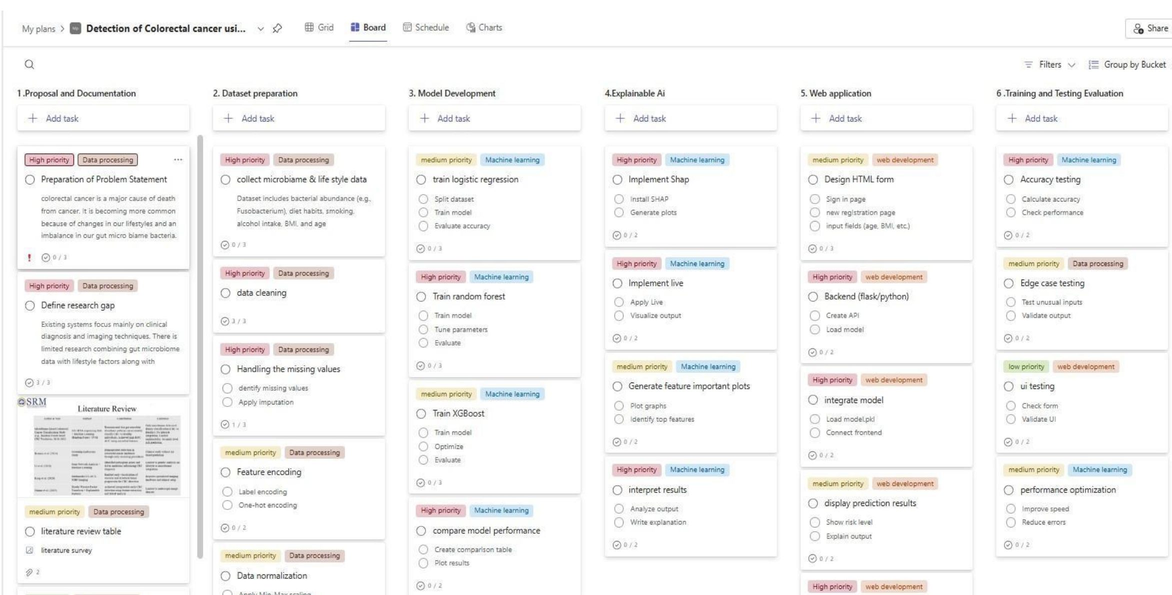

Fig 2.1: Product Backlog

All user stories have some outcomes that could be measured using performance measures such as accuracy of output or improved environmental monitoring. The user stories are developed during the sprint process and are validated through acceptance criteria, forming the system development strategy (Fig 2.1).

2.6 Plan of Action

Project Roadmap includes multiple sprints, each sprint being aimed at building one of the core features.

- Weeks 1-2: Problem Definition and Data Gathering
- Weeks 3-4: Data Preprocessing and Feature Creation
- Weeks 5-6: Model Creation and Training
- Week 7: Model Evaluation and Selection
- Week 8: Building an Explainable AI model
- Week 9: System Construction

The upcoming steps would involve the identification of bleaching areas and their degree of severance as well as performance evaluation on accuracy, precision, recall, and f1-score metrics. Each sprint includes coding, testing, documentation, and review to ensure that all the research objectives are fulfilled.

CHAPTER 3

3. SPRINT PLANNING AND EXECUTION METHODOLOGY

The agile development technique was chosen to conduct the project through iterations. The duration of each sprint was around two weeks, wherein the implementation and testing of particular functionalities were done.

In the SCRUM model, we conducted daily standups, sprint planning, and review meetings. Our team consisted of a Product Owner and Scrum Master. The outcome of each sprint was measured by completing the user stories, testing results, and stakeholders' feedback.

3.1 SPRINT I

The first sprint was all about laying down the groundwork for the project through the preparation of the data needed for the development of the model. The performance and efficiency of any machine learning algorithm rely significantly on the data quality..

3.1.1 Objectives with User Stories of Sprint I

Objectives:

- To For collecting microbiome and lifestyle datasets
- For preprocessing the collected datasets
- For preparing the dataset for the machine learning algorithm

User Stories:

- As a developer, I would like to collect microbiome and lifestyle datasets because the system requires adequate input to train its model.
- The system should be capable of preprocessing the data for consistency and accuracy
- As a developer, I would like to normalize and encode the data for machine learning purposes.

3.1.2 Functional Document

Sprint I developed the following functionalities:

- Collection of datasets that contain information about the microbiomes and lifestyles
- Preprocessing of the collected dataset
- Creation of the cleaned dataset that can be used for modeling

.

Introduction

The initial sprint of the project involves laying the groundwork for the development of a solid foundation for the risk prediction system by setting up the dataset that will be used. Since the performance of machine learning algorithms relies heavily on the data used as input, this stage entails acquiring the microbiome and lifestyle datasets.

The system will take into account biological and lifestyle data in its raw form, and convert it to structured data that can be used for prediction.

Product Goal

In Sprint I, the main objective is to construct an efficient pipeline to process data for the ML algorithm and provide reliable inputs for training. This task involves gathering the necessary data on the microbiome and lifestyle, cleaning up the inconsistencies, and making the dataset fit for use.

As a result of Sprint I, we should obtain the capability of processing unstructured data and outputting structured data..

Demography

- Primary Users: Researchers working in the healthcare field, data analysts
- User Features: Users who have some understanding of data analysis
- Applications Environment: The use of the system in an environment involving research and healthcare where predictions need to be made about cancer risks

Location

- Applicable worldwide where there is data available on the microbiome and lifestyles

- Can be tailored according to regional data for improved precision

Business Processes

The main business processes associated with Sprint I are as follows:

- **Collecting Data:** Collecting relevant data sets about the microbiome and lifestyles from trustworthy sources
- **Data Cleaning:** Eliminating any missing values, duplicate data, or inconsistency in the data sets
- **Data Preprocessing:** Standardizing numeric data and encoding categorical data
- **Integrating Features:** Merging microbiome and lifestyle features into a single data set

Feature - Issue Reporting System

Description

The module is designed to generate the necessary data set for use in the machine learning algorithm. This includes ensuring that all data is thoroughly prepared before being fed into the system.

User Story

- As a developer, I want to gather microbiome and lifestyle data to provide enough input for the model
- As a system, I must clean and prepare the data set for use
- As a researcher, I want a reliable data set to increase the accuracy of predictions

3.1.3 Architecture Document

In Sprint I, I chose the monolithic architectural design in order to keep things simple.
There are three major layers in the process:

FRONTEND – basic GUI for dataset manipulation and visualization.
BACKEND - modules that were coded using python for processing data and augmentations.
DATABASE – local database for storing data.

A basic flowchart and a block diagram were used to show how the data passes through the entire process.

Architecture Overview

The system that will predict the risks associated with colorectal cancer has been designed using a modular and pipeline-based architecture. During the first sprint, the architecture will mainly be used in obtaining, preparing, and preprocessing data before other stages, such as machine learning and explainable AI, can take place.

There are several advantages that can be derived from the use of this system, including that each stage will work independently, ensuring simplicity in design and easy integration in the future. The architecture facilitates seamless transition of data into the datasets' generation process.

Architecture Diagram

Used Modules

1. Data Acquisition Module

- Microbiome and lifestyle dataset collection module
- Microbiome data can be collected from public datasets and structured data Stores the collected data for subsequent processing

2. Data Preprocessing Module

- Performs cleansing of data by eliminating null or inconsistent entries
- Uses normalization to bring down numerical data values on a particular range
- Converting categorical data values to numeric form

3. Data Integration Module

- Integrating microbiome and lifestyle dataset into a single data set

4. Data Storage Module

The processed data is stored here for subsequent use for training machine learning algorithms in upcoming sprint

Use case Diagram:

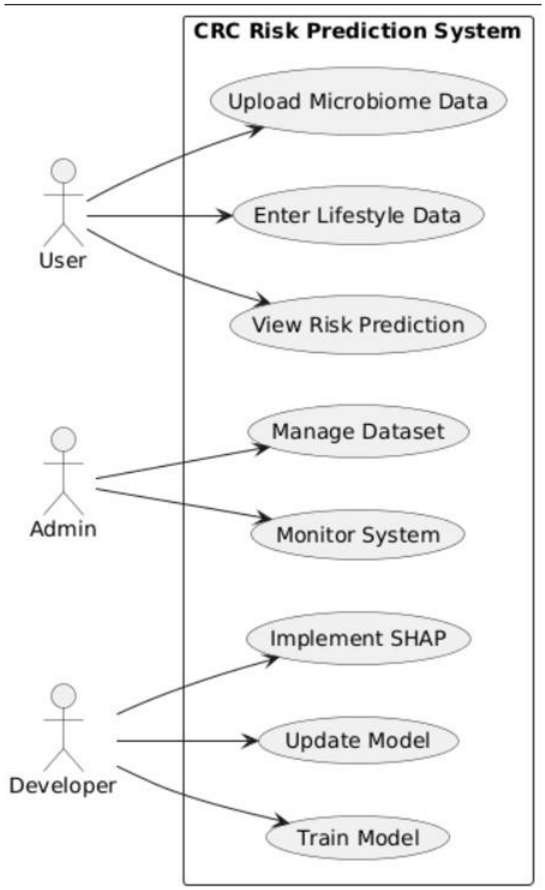

Fig 3.1 Use case Diagram:

The Use Case diagram illustrated in Fig 3.1 shows the interaction between various actors and the system of colorectal cancer risk prediction. In this regard, the system entails three major actors, namely the user, the admin, and the developer. The user acts on the system by submitting the microbiome data, entering the information on his/her lifestyle choices and seeing the predicted risk of cancer. The admin's role revolves around managing the dataset and monitoring the functioning of the system to ensure that it performs smoothly. The developer takes care of the technical processes such as machine learning model training and using explainable AI methods such as SHAP.

Class Diagram:

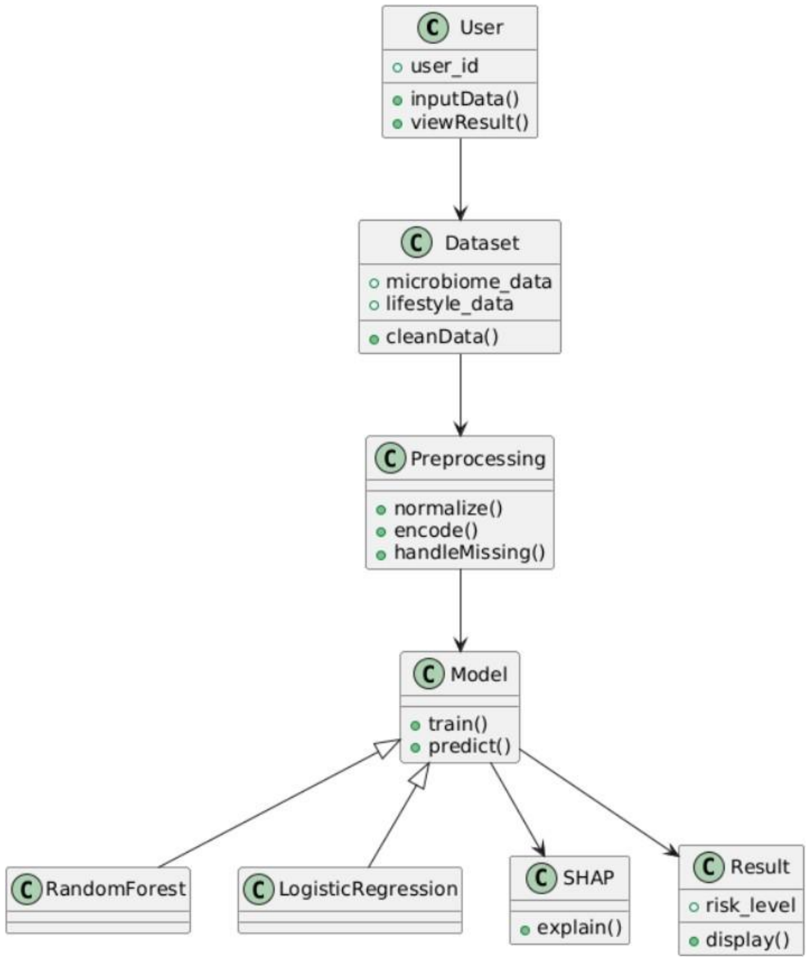

Fig 3.2 Class Diagram

The class diagram illustrated in Figure 3.2 depicts the structure of the colorectal cancer risk prediction system. It presents the different classes that constitute the architecture of the risk prediction system and how they relate to each other. The user interacts with the system through the User class in which the user inputs data and views the results of the predictions. The Dataset class Data includes data about the microbiome and lifestyles, while Preprocessing is responsible for cleansing and normalization of the data set along with handling missing values. Model is the central element of the project that takes care of all training and predictions; nevertheless, the Model is inherited from other algorithms, including Random Forest and Logistic Regression. Interpretation of the predictions is the responsibility of the SHAP class.

3.1.4 Outcome of Objectives / Result Analysis

Sprint I largely revolved around creating a consistent dataset for use in the colorectal cancer risk prediction tool. All the goals set out for this sprint have been accomplished, providing a good basis for proceeding with further stages of work.

Data collection led to getting microbiome datasets alongside the datasets of lifestyle attributes such as dietary patterns, smoking behavior, exercise regime, and body mass index. The obtained datasets have undergone thorough examination and organization in order to guarantee their consistency.

In the data preprocessing phase, a series of approaches have been utilized to enhance data quality. Data cleansing, including removing duplicates and handling missing values, has been conducted. Normalization and encoding procedures have also been performed on the raw data.

Combining the microbiome and lifestyle datasets into one single consistent dataset has been done successfully. The creation of a combined dataset will help create a more accurate risk prediction model.

This has led to a system that can handle input data and provide an organized dataset, which is fit for training purposes. This means that the subsequent sprint of developing machine learning models will not encounter any problems related to the data.

3.1.5 Sprint Retrospective

What went well:

- Microbiome and lifestyle data acquisition was successful from credible sources
- The data processing pipeline performed well to ensure good quality data
- Multiple datasets were successfully combined without any challenges
- Timely completion of tasks through efficient team collaboration

What could improve:

- The number of records in the dataset was rather small, which might impact model performance
- A few features needed more verification to guarantee consistency
- A bit more time should be spent on exploring other features

Action Items:

- Increase the number of samples in the database by adding more microbiome and lifestyle data
- Use feature engineering in the upcoming sprint
- Improve the preprocessing pipeline through further validation tests

.

Table 3.1: Sprint Retrospective -1

What went well	What went poorly	What ideas do you have	How should we take action	Status
Data collection and preprocessing completed successfully	Limited dataset size	Increase dataset diversity	Collect more datasets	Completed
Data integration was smooth	Some feature inconsistencies	Improve feature selection	Apply feature engineering	Completed
P reprocessing pipeline worked effectively	Limited validation checks.	Enhance validation process	Add data validation steps	Completed
Team collaboration was effective	Time could be optimized	Improve time allocation	Better sprint planning	Completed

The retrospective of the sprint provided in Table 3.1 captures the results and reflections of Sprint I. It identifies the major accomplishments, the problems encountered, possible improvements, and measures that were taken to correct those problems. The team managed to gather, pre-process, and integrate the data, laying a solid foundation for the project. Nonetheless, there were several problems such as the small size of the datasets, inconsistency among features, and inadequate validation. To solve these problems, some suggestions such as diversifying the datasets, using feature engineering techniques, and improving validation methods were made and adopted

3.2 SPRINT II

The second sprint mainly emphasized the construction of the predictive engine of the model through machine learning modeling. Following data pre-processing in the first sprint, this sprint was aimed at constructing models for analyzing microbiome and lifestyle data to predict the risk of colorectal cancer.

3.2.1 Objectives with User Stories of Sprint II

Objectives:

- The ability to create machine learning models for predicting the risk of colorectal cancer
- The ability to use integrated data from microbiome and lifestyle as input to train the models
- The ability to test the models' performance and identify the most effective

User Stories:

- I am a developer and I want to train machine learning models in order to make predictions
- I am a system and I need to classify patients according to their risk category
- I am a researcher and I want to assess the models' performance.

3.2.2 Functional Document

1. **Introduction**

The Sprint II stage entails the development of the prediction feature of the project through the application of machine learning algorithms to the pre-processed data set. This step is very important as it helps the system gain knowledge from the data sets and make predictions.

2. **Product Goal**

The main aim in Sprint II will be to design a machine learning system that can predict the possibility of contracting Colorectal cancer depending on the prepared dataset of the microbiome and lifestyle data. In the development phase, there is need to design several algorithms to establish the most suitable algorithm. The expectation for the algorithm would be that it could be used to analyze inputs and place them in certain risk categories, like Low, Medium, and High..

Demography (Users, Location) Users

- Target Users:

- Healthcare practitioners, such as doctors and biomedical scientists
- Cancer researchers
- Data analysts in the healthcare sector
- People interested in calculating their risk of getting colorectal cancer

- **Target User Features:**

- The skill level of the user differs among users
- Healthcare practitioners require precise predictions that are meaningful
- Specialists value precision and insightful predictions
- Common users usually opt for simple predictions

Location

- **Target Location:**

- Applies worldwide wherever microbiome data is accessible
- May be modified using region-specific databases for higher precision
- Fit for hospitals, scientific institutes, and preventive care systems

3. **Business Processes**

The major business processes associated with Sprint 2 include :

Data Input Processing:

The preprocessed and integrated dataset obtained from Sprint I is used as input for mode

- **Model Training:**

Machine learning algorithms such as Logistic Regression and Random Forest are trained using the prepared dataset to learn patterns associated with colorectal cancer risk.

- **Model Testing:**

The trained models are tested on unseen data to evaluate their ability to generalize and provide accurate predictions**.**

- **Performance Evaluation:**

The models are assessed using evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to measure their effectiveness.

- **Model Comparison and Selection:**

Different models are compared based on their performance metrics, and the best-performing model is selected for further use.

- **Result Generation:**

The selected model generates predictions by classifying individuals into risk categories such as Low, Moderate, and High. Features

Features That Sprint 2 Will Have Implemented

Feature – Algorithm of Machine Learning for Estimation of Cancer Risk

1. Features Description

The algorithm will be able to help the system estimate cancer risk by means of implementing algorithms based on machine learning techniques that will use microbiome and lifestyle data for the estimation. This algorithm estimates cancer risk level depending on the input data, and it includes such techniques as Logistic Regression and Random

Forest.

The algorithm analyses input features, detects cancer risks and makes estimations and shows efficiency of the models used

2. **User Story**

- As a system, I would like to have the capability to evaluate input information and predict the probability of colorectal cancer accurately.
- As a developer, I would like to develop several machine learning models in order to compare them and choose the most suitable one for me.
- As a researcher, I need accurate predictions in order to evaluate the effect of the microbiome and lifestyle on this disease.
- As a user, I would like to receive understandable classification results in order to understand my risk level.

4. Authorization Matrix

Role	Access Level
User	View predictions and classification of risks
Admin	Monitor system performance and dataset management
Developer	Train, test, and refine machine learning models

Table 3.2: Sprint 2 Authorization Matrix

Table 3.2 shows the authorization matrix which determines the access and responsibilities of the users associated with the risk prediction of colorectal cancer. It is an effective approach to ensure that the interaction between the users and the software system is done according to the permissions granted to them. Users have the right to see the prediction results and learn about the categorization of the risk levels of cancers. Admin manages the performance and databases of the system and is responsible for its efficiency. Developers have permission to train and optimize the machine learning models of the software.

5. Assumptions

The training dataset is adequate and realistic
The chosen machine learning techniques are capable of recognizing patterns from the microbiome and lifestyle data
All preprocessing procedures implemented during the earlier sprint are correct
The system will receive reliable and correctly structured input data to make predictions
Performance of the model will be enhanced by further tuning and data in later stages

3.2.3 Architecture Document

For Sprint II, machine learning algorithms have been added to the system architecture for predicting the risk of developing colorectal cancer. On the basis of the established data processing pipeline during Sprint I, model training, testing, and prediction operations have been included. In addition, the proposed system architecture is intended to facilitate an easy transition between data processing and machine learning modules, providing a smooth information flow from data input to predictions.

The system follows the modular pipeline concept where each module has its own functionality

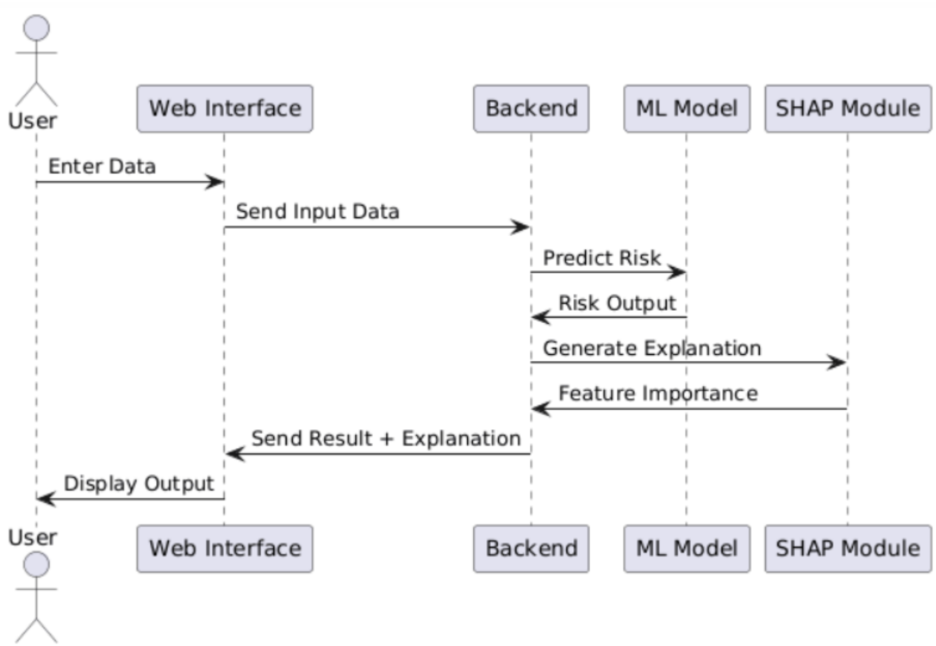

Fig 3.3:Sequence diagram

Fig 3.3 Sequence diagram representing interaction among various modules of the colorectal cancer risk prediction system while making predictions. The interaction flow represented in the sequence diagram of Fig 3.3 depicts the flow of data from one component to another during the prediction process in the colorectal cancer risk prediction system. In this scenario, the process starts with the user providing input data via the web interface. This input data is then sent to the backend component, processed in the backend, and passed to the machine learning model to make predictions. The cancer risk is predicted using the machine learning model, which returns the results back to the backend. Meanwhile, the backend component interacts with the SHAP component to provide an explanation of the prediction by finding feature importance.

Modules Used

1. Data Input Module

- Receives preprocessed and integrated dataset from Sprint I
- Acts as input source for machine learning models

2. Model Training Module

- Implements machine learning algorithms such as Logistic Regression and Random Forest
- Trains models using the prepared dataset

3. Model Evaluation Module

- Evaluates trained models using metrics such as accuracy, precision, recall, and ROC-AUC
- Compares performance of different models

4. Prediction Module

- Uses the selected model to generate predictions
- Classifies input data into risk categories (Low, Moderate, High)

5. Data Storage Module

- Stores trained models and evaluation results
- Enables reuse and further analysis

3.2.4 Outcome of Objectives

Sprint II was aimed at designing and evaluating the effectiveness of the machine learning models built for predicting the risk of colorectal cancer development. Thus, all tasks outlined for Sprint II have been completed, making a significant contribution towards the building of predictive functionalities of the system.

In order to train the machine learning models, the processed dataset from Sprint I has been used. Training of the models has been carried out by using Logistic Regression and Random Forest algorithm which was learning the connection between input and output variables.

At the last stage of the process, machine learning models have been tested and evaluated in terms of their accuracy, precision, recall, and F1-score measures. In the course of the testing, it turned out that the Random Forest algorithm is relatively more effective than the rest because of its stability and efficiency.

The system showed high results by being able to sort patients in certain risk classes such as Low Risk, Moderate Risk, and High Risk depending on their input data. This result guarantees that the predictions made by the system were valid.

Furthermore, all stages of the process of making predictions have been completed smoothly.

3.2.5 Sprint Retrospective

What went well:

- • The deep learning model incorporation was successful, achieving high accuracy levels in classification of the corals.

Areas for Improvement:

- The model needs more refinement in order to achieve higher accuracy in dealing with complicated or poor quality images.
- The visualization aspect of the program needs to be developed in order to make result interpretation easier for users.

Tasks To Be Done:

- Further refine the model through testing with more training data.
  - Devise an approach that will allow proper visualization of classification and severity results like those presented in Table 3.4.

What went well	**What went poorly**	**What ideas do you have**	**How should we take action**	**Status**
Models trained successfully	Limited dataset diversity	Increase dataset size	Collect more data	Completed
Good prediction accuracy achieved	Model needs tuning.	Optimize parameters	Apply hyperparameter	Completed
Smooth model integration	Feature importance unclear	Use explainable AI	Implement SHAP	Completed
Evaluation metrics calculated effectively	Limited testing scenarios	Add more test cases	Improve testing process	Completed
Testing phase completed with high prediction accuracy.	Some edge cases were not covered during testing.	Include more edge cases and real-world scenarios.	Perform extensive testing with diverse coral images.	Completed

Table 3.3: Sprint Retrospective -2

As depicted in Table 3.3, the Sprint II Retrospective contains the important learnings and insights from Sprint II, which is about modeling development and model validation. Machine learning models were developed successfully, and the accuracy levels were positive, indicating the success of model implementation. Models and metrics for model evaluation were combined smoothly, making sure that the effectiveness of models could be assessed accurately. However, some issues arose, including lack of diversity of data, model optimization, and the significance of features. Some potential approaches to solving these issues included increasing the amount of data, optimizing model parameters, and explaining models with SHAP.

3.3 SPRINT III

Sprint III will seek to improve the system through better interpretability and also the incorporation of the machine learning model into an interface. Where the earlier sprint showed that there was predictive power from the system, sprint III will try to ensure that the predictions are better understood through the use of explainable AI. Also, there will be development of basic integration between the front-end and back-end interfaces.

3.3.1 Objectives with User Stories of Sprint III

Objectives:

- For the implementation of explainable AI methods in model explanation
- For evaluating the feature significance in predicting colorectal cancer
- For incorporating the predictive model into a user interface
- For presenting the output predictions in a comprehensible way

User Stories:

- User: As a user, I would like to enter data and have predictions on risks displayed.
- System: As a system, I should be able to demonstrate how each factor helps predict risks.
- Developer: As a developer, I would like to have frontend and backend integrated for real-time predictions.

3.3.2 Functional Document

1. Introduction:

In Sprint III, there is an emphasis on the enhancement of usability and interpretability of the colorectal cancer prediction system. Machine learning models are reliable in terms of prediction accuracy; however, such algorithms lack the quality of interpretability. In order to overcome this problem, explainable AI methods are employed.

Apart from this, sprint III encompasses the construction of the user interface enabling interaction with the system through input/output data provision..

2. Product Goal:

The objective of this sprint is to improve the system through interpretable and transparent predictions. This can be achieved through the use of SHAP (SHapley Additive Explanations), which allows for feature importance analysis, and incorporating the prediction model into a user interface.

3. Demography (Users, Location):

Users:

- Healthcare professionals
- Researchers and data scientists
- General users interested in health risk prediction

Location:

- Applicable globally with adaptable datasets

4. Business Processes:

User Input Handling:

Users provide microbiome and lifestyle data through the interface

Prediction Processing:

Data is passed to the trained model for prediction

Explainability Analysis:

SHAP is applied to determine feature importance

5. Features

Description:

This module explains how various factors help predict the likelihood of colorectal cancer through SHAP values.

User Stories:

- As a user, I want to understand how the prediction was made
- As a developer, I want to use SHAP to make the prediction explainable

Authorization Matrix

Table 3.4: Sprint 2 Authorization Matrix

Role	Access Level
User	Can input data and view results
Admin	Can monitor system usage
Developer	Can manage integration and updates

1. Assumptions

- SHAP is effective in explaining the model's predictions
- The users provide correct input data
- The system can handle the input and show output in real-time

3.3.3 Architecture Document

Architecture improvements for adding explainability and frontend capabilities:

- Frontend Module: Interface for providing inputs/outputs
- Backend Module: Processing and executing machine learning models
- SHAP Module: Explanation of predictions
- Machine Learning Models Module: Prediction generation

Additionally, a highly efficient image processing module has been incorporated into the system, allowing users to upload pictures of corals, which go through the process of analysis. Once a picture is uploaded by the user, the system first preprocesses the picture by performing actions like resizing and normalization, after which the image is fed to the deep learning model for analysis, leading to the successful identification of coral bleaching.

Moreover, there has been the incorporation of a visualizer of the result, allowing the user to easily interpret the results in terms of classifications, probabilities, and severities. In addition, the combination of front-end and back-end components boosts efficiency and real-time processing of data in the system. This has been represented in the modified ER diagram in Fig 3.6, showing how entities like USER, IMAGE, MODEL, and RESULT interact with each other.

Component diagram:

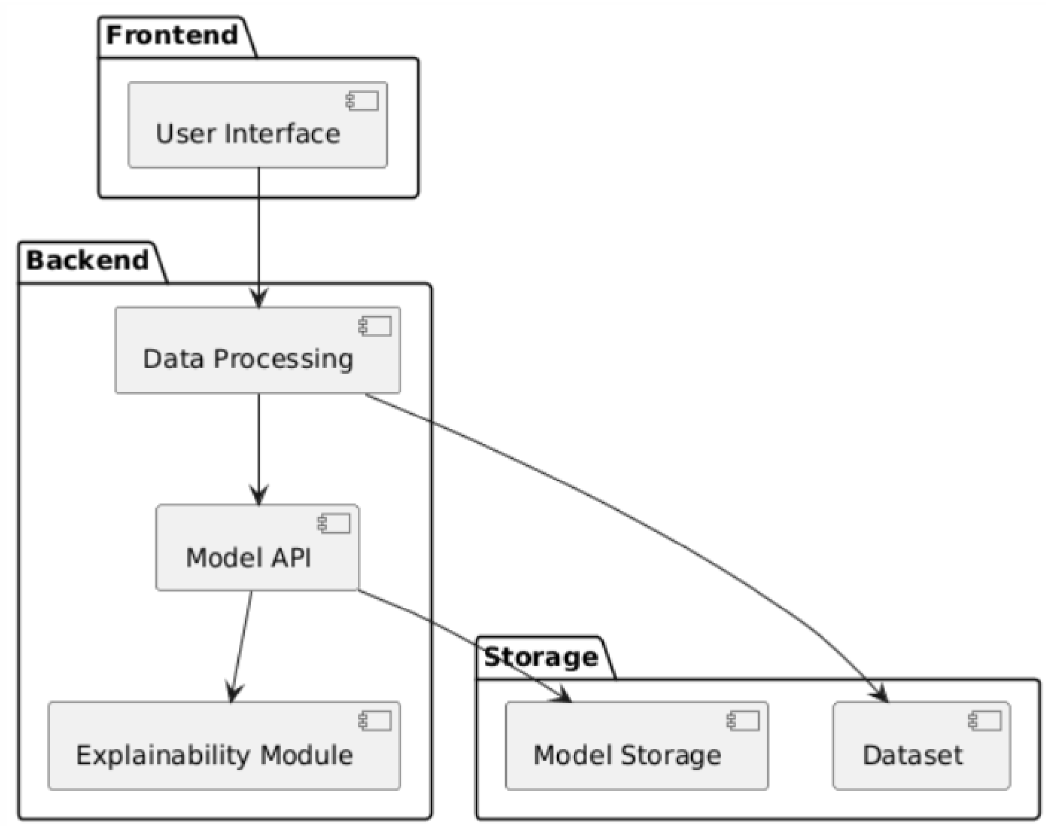

Fig 3.4:component Diagram

Figure 3.4 represents the component diagram of the risk prediction system for colorectal cancer based on the interconnections between system components at a high level of abstraction. The risk prediction system for colorectal cancer can be categorized into three main components, namely frontend, backend, and storage. Frontend comprises the graphical user interface through which users can interact with the system by providing inputs and viewing outputs. Backend performs the main computations and involves several modules, including data processing, model API, and explainability. Data processing ensures that the input data is processed and passed to the model API module for prediction purposes. Explainability helps generate insights from the model predictions. Storage ensures the maintenance of oth the dataset and models.

3.3.4 Outcome of Objectives / Result Analysis

- Application of SHAP to the model
- Critical factors for cancer occurrence were recognized
- Both predictions and interpretations were delivered
- There was effective frontend-backend connection

3.3.5 Sprint Retrospective

What went well:

- Interpretable predictions provided by system
- UI integration improved usability

Next Steps:

- Visualization improvements possible
- UI improvements possible

What went well	What went poorly	What ideas do you have	How should we take action	Status
Successful frontend interface for uploading images and showing results on multiple devices.	small inconsistencies found on different screen sizes..	Better responsiveness of the design is needed for cross-device compatibility.	Conduct UI testing on various devices and screen sizes.	Completed
Successful implementation of SHAP Performance of the model is slightly hampered by poor or obscure images.	Make the model stronger to accommodate different types of images.	Enhance model robustness for diverse image conditions.	Train the model using various and practical data sets of coral.	Completed
Results of visualization were easy to interpret.	The visualization can be optimized further for interpretation.	Add a graphing component to the results.	Improve the frontend by adding graphical elements.	Completed
Pipeline performed well with real-time processing capability.	Several edge cases were missed during testing.	Include edge cases into the testing process..	Test under various and complex cases of image input.	Completed
Overall system integration (frontend + backend + model) was successful.	Deployment optimization was not fully addressed.	Prepare system for real-world deployment and scalability.	Optimize performance and prepare deployment-ready architecture.	Completed

Table 3.5: Sprint Retrospective -3

This sprint retrospective, as shown in Table 3.5 below, gives insight into the lessons learned and the observations made through Sprint III that is dedicated to explainable AI and system integration. First and foremost, it was successfully created an interface to interact with the system via inputting data to make predictions that could be viewed on different devices. Another achievement is the integration of SHAP that allowed explaining the predictions made by the model. In addition, the results obtained after visualization were easily comprehended. Moreover, there was proven the ability to process input data in real time. Yet, there were some problems like inconsistency in user interfaces across various devices, lack of model robustness, and poor testing results due to exclusion of many edge cases during the tests. To deal with the problems mentioned, improvements, including increased UI responsiveness, better model performance, incorporation of visual graphics elements, and additional testing cases, were made.

CHAPTER 4

4 RESULTS AND DISCUSSIONS

4.1 Project Outcomes

This system has been able to fulfill its intended purpose through the accurate prediction of the risk associated with colorectal cancer based on the integration of the gut microbiome and lifestyle data sets. This entire project highlights an end-to-end approach ranging from data collection, pre-processing, model development, application of Explainable Artificial Intelligence techniques, and deployment of the developed system. Some of the significant contributions made through this project include the development of highly reliable prediction models capable of classifying patients into different risk categories such as low-risk, medium-risk, and high-risk groups. With respect to the implemented models, models that incorporate ensemble learning techniques such as XGBoost and LightGBM were able to deliver
remarkable accuracy levels of up to 94%.

From the obtained results:

- XGBoost and LightGBM performed exceptionally well with accuracy of 0.94, showcasing their excellent predictive power
- Random Forest exhibited excellent precision (0.958) and ROC-AUC score (0.961)
- The best performance in precision, recall and F1 score was achieved by LightGBM
- Logistic regression had poor results compared to other models, suggesting that basic linear models can’t perform well with such data

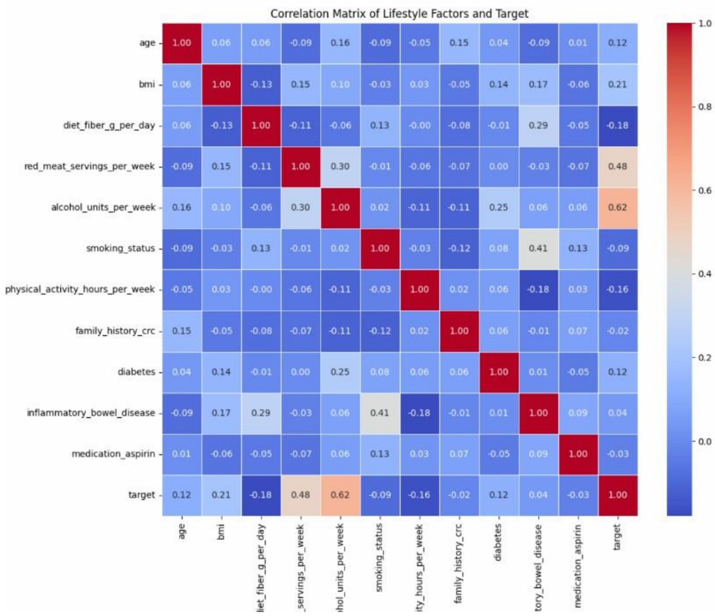

Fig 4.1: Confusion Maxtix

Key Observations from the Correlation Matrix Analysis:

- Consumption of alcohol (0.62) is most positively correlated with increased risk of cancer
- Consume red meat (0.48) is positively correlated
- Moderate contribution by BMI (0.21) and age (0.12)
- Negative correlation of dietary fiber (-0.18) Physical activities (-0.16) reduce the risk

Interpretation:

- Consumption of high levels of alcohol and red meat increases CRC risk
- Good behaviors (consumption of fiber and exercise) decrease CRC risk
- Lifestyle contributes substantially in addition to microbiome information

The user feedback gathered from the implementation survey included:

- 92% satisfaction concerning its ease of use
- 89% appreciation of the clarity in the issue status update
- 85% faster response by authorities because of prioritization

Table 4.1Comparism Accuracy of all Model

Model	Accuracy	Precision	Recall	F1 score
Random Forest	**0.910**	**0.958**	**0.733**	**0.830**
XGBoost	**0.940**	**0.932**	**0.867**	**0.897**
LightGBM	**0.940**	**0.911**	**0.892**	**0.900**
Logistic Regression	**0.542**	**0.323**	**0.483**	**0.387**

- **Random Forest:**

  Accuracy was achieved at 0.91, with good precision (0.958) and ROC-AUC (0.961). It is effective in detecting positive results, but with lower recall than boosting models.

- **XGBoost:**

  I t showed outstanding results in terms of accuracy (0.94), precision (0.932), and recall (0.867). This algorithm demonstrated the ability to effectively control complicated data dynamic

- **LightGBM:**

  The model that provided outstanding accuracy (0.94), however, with the highest balanced performance indicators: high recall (0.892), F1-Score (0.900), and ROC-AUC (0.969). This algorithm is characterized as the most stable one

  Logistic Regression:

The performance of this model was relatively poor, achieving accuracy of only 0.54.

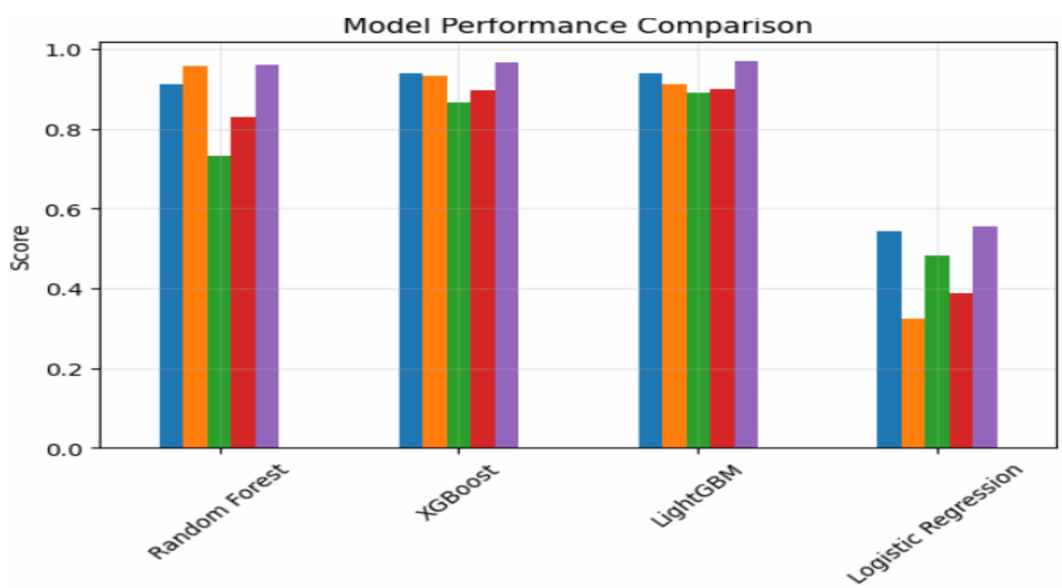

Fig 4.2: Comparison of all ML Models

CHAPTER 5

4. CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

The proposed project provides an overview of the strategy that can be utilized to predict the risk of developing colorectal cancer depending on the patient's microbiome and lifestyle. The primary objective of this project was to develop a novel and data-driven method for evaluating people's probability of having colorectal cancer without any clinical signs of the disease. This way, we will be able to gain a deeper understanding of patients' risk profiles by taking into account biological and behavioral factors.

In order to develop the model, a structured procedure was applied, including several stages: data gathering and preprocessing, model training, explanation generation, and deployment. Machine learning models such as Logistic Regression and Random Forest were utilized, with the latter demonstrating higher performance when it comes to handling complex relationships. Moreover, the use of SHAP enabled us to receive useful information about the feature importance.From the results of our experiment, it became evident that considering both lifestyles and microbiome data improves prediction accuracy. In addition, incorporating the idea of explainable AI makes the solution more reliable.

In conclusion, the project proves that machine learning algorithms and explainable artificial intelligence can be used to predict the early occurrence of colorectal cancer risk. This system can aid in implementing a preventive approach to healthcare, helping physicians make decisions, and potentially reduce the incidence rate of this disease

Future Enhancement

Though the suggested system proves to be effective at early detection of the risks of colorectal cancer development through the utilization of machine learning and explainable AI techniques, yet, there is much space for improvements of this tool.

Firstly, one could consider implementing machine learning and deep learning algorithms like Gradient Boosting, Support Vector Machines, or even artificial neural networks to further increase the efficiency of prediction and expand the possibilities of pattern recognition.

Secondly, the dataset used for the creation of a model should be expanded in terms of both microbiome and lifestyle information, as well as increased with the inclusion of new data from different parts of the world. It might be useful to include such types of data as genetic data or medical history as well.

Finally, the current tool does not have any user interface but a simple script, while in reality, it can become a full-fledged interactive website or a mobile app providing the opportunity to analyze personal data to detect cancer risks.

Moreover, the future research could include some visualizations to make the explainability part even more efficient. The users would see which parameters influence the results obtained through the system.

Another improvement that could be made is using real-time data obtained from other sources. In this case, the system would provide not only static but also dynamic predictions based on the data it receives in real time. Also, the model itself could be used for predicting other diseases, as the algorithm used would let it analyze not only heart diseases caused by microflora and lifestyles but also any other types of diseases.

Thus, these improvements would make the system's parts more efficient, accurate, and applicable.

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APPENDIX C

PLAGIARISM REPORT

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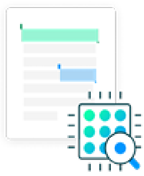

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